



AURATONE AI

K I - G E S T Ü T Z T E A U D I O M A S T E R I N G W O R K S
T A T I O N

v2.0.0 – Standalone macOS Release (Apple Silicon)

by KKEEY · Kevin Kuck · Lead System Architect & AI Engineer · August 2026

24

MODULE

18k+

LINES OF CODE

48

TESTS

32

TAURI COMMANDS

94 MB

DMG BUNDLE

Systemarchitektur & Tech Stack

Tauri 2.0 Single-Binary Architecture: React 19 Frontend, Rust Tokio Kernel, Metal GPU & Hermetic DSP

FRONTEND – BROWSER & REACT 19

React 19

TypeScript 5.8

Vite 6.4

Tailwind CSS 4

Motion (Framer)

Metal GPU Canvas

Recharts

Lucide Icons

Web Audio API

IndexedDB

↓↓↓

KERNEL – TAURI 2.0 RUST CORE

Rust 1.80+

Tokio Async Runtime

DspSupervisor

3-Way SQLite Triade

FSEvents Watcher

macOS Keychain

Apple Music JXA Bridge

↓↓↓

DSP ENGINE & HARDWARE – NATIVE SIDECARS & GPU

dsp-aarch64-apple-darwin (Python/Librosa)

ffmpeg (LUFS Engine)

metal_peaks.rs (Metal Compute)

Unified Memory

↓↓↓

CLOUD & INTEGRATIONEN

GCP Cloud Run (Mastering Fallback)

Google Cloud Storage

Gemini 1.5 Pro

Apple Music.app

Traktor Pro 4

DATENHALTUNG – 3-WAY SQLITE TRIADE

catalog.db

Track-Metadaten, BPM, Camelot Keys, Energy Scores & Traktor Marker ID Matchings.

cache.db

Metal Peak-Pyramiden, FFT Spectral Data, Audio Blobs & Pre-rendered Waveform Cache.

history.db

Atomic Audit Logs, Pipeline Job Queue State & Crash Recovery Execution Snapshots.

Audio Engine & DJ-Workstation

Die Kernfunktionalität: Native Waveforms, Mastering, Key/BPM Analyse, Curation und Traktor Export

AuraMaster — Mastering Engine

PRODUKTIV

- FFT-Spektrum-Analyzer (44.1 kHz, High-Res Buffer)
- Automatische LUFS-Normalisierung (-12 LUFS Target)
- 3 Modi: Power Block, Sanft Harmonisch, Peak Focus
- Hermetisches FFmpeg Sidecar (Echte Filterkette)
- Output: key_bpm_songname_artist.wav

AuraHarmonics — Key/BPM & Camelot

PRODUKTIV

- Hermetisches Python DSP Binary (librosa Engine)
- Camelot Harmonic Wheel (1A–12B) Visualisierung
- Harmonisches Set: Automatische Camelot-Sortierung
- Übergangs-Scoring zwischen Tracks
- Traktor-Theme-Farben für Wheel und Decks

Dual-Deck DJ Player

PRODUKTIV

- Deck A/B mit Metal-GPU Waveform Canvas
- Crossfader mit Echtzeit-Mixing Engine
- BPM-Sync & Scratch/Pitch Controls
- ORIG / MASTER Quellmodus-Umschaltung
- Cue-Point Management (Add/Edit/Remove)

AuraCurator — KI-Playlist

KI / GEMINI

- Gemini 1.5 Pro als Musikologe & Archivar
- KI-Playlist-Sortierung nach Energie & Harmonie
- Transition-Tipps & Harmonic Compatibility
- Beat-Sync FX: FLTR, STTR, DLAY, RVRB
- Dynamische Energie-Kurve (Intro/Main/Outro)

AuraMetal — GPU Waveform Engine

NEU
V2.0

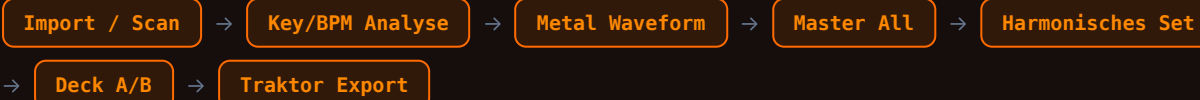
- Apple Silicon Metal Compute Shader Pipeline
- Direct Unified Memory Peak Pyramid Calculation
- Null CPU Overhead bei Zoom & Waveform Scrolling
- Automatischer Fallback auf Rust Multithread CPU
- Integrations-Brücke zu React Canvas Component

AuraTraktor & Apple Music

PRODUKTIV

- Import: Drag&Drop, Ordner-Scan, JSON, NML
- Export: .nml, .m3u, .csv (Traktor Pro 4 Ready)
- Apple Music Integration via JXA Bridge (osascript)
- Pfad-Auflösung bevorzugt masteredPath
- Real-time Directory Watching (FSEvents)

WORKFLOW — GIG-VORBEREITUNG



10-Punkte Standalone macOS Release (v2.0.0)

Alle Architekturelemente und Systeme des vollendeten v2.0.0 Standalone Desktop Releases

Error Handling & DspSupervisor

01

Rust DspSupervisor kontrolliert Sidecar-Subprozesse mit Timeout, Cancellation Tokens und Fehler-Recovery.

`dsp_supervisor.rs` · `ErrorBoundary.tsx` · `ToastSystem.tsx`

Benutzerprofile & Targets

02

LUFS-Target, Declip, HPF pro Profil konfigurierbar. Isoliertes SQLite-Storage, sofort aktiv.

`profileStore.ts` · `catalog.db` · `loadActiveProfile()`

Analytics & Native Diagnostics

03

Strikt lokales Client-Logging & Server metrics. Request-Timing, Latenzen & Audit-Logs in history.db.

`analytics.ts` · `history.db` · `get_audit_log()`

Hermetische Sidecar Pipeline

04

Single-Call Ingest Pipeline: Upload → DSP-Binary → FFmpeg Master → SQLite Update in einem Durchlauf.

`pipeline.rs` · `dsp-aarch64-apple-darwin` · `ffmpeg`

Batch Processing & Tokio Pool

05

Parallele Multi-Track-Verarbeitung mit Tokio Worker-Pool (Concurrency Controlled) & Abort-Signal Handling.

`batch.rs` · `Tokio Tasks` · `CancellationToken`

Onboarding & Hilfe System

06

Interactive 6-Schritt Tour für Erstbenutzer. Hilfe-Panel mit Shortcuts-Referenz & FAQ-Modal.

`OnboardingOverlay.tsx` · `HelpPanel.tsx` · `onboardingStore.ts`

Bundle-Optimierung

07

Vite 6 Code-Splitting: 6 Vendor-Chunks (react, recharts, motion, firebase, audio, icons). Ultra-fast Loading.

`vite.config.ts` · `manualChunks` · `React.lazy()`

macOS System Integration

08

Apple Keychain verschlüsselte Key-Verwaltung, FSEvents Verzeichnis-Watcher & JXA Apple Music Sync.

`keychain.rs` · `watcher.rs` · `apple_music.rs`

Cloud Mastering Fallback

09

Asynchroner GCP Cloud Run Fallback Microservice für rechenintensive Mastering-Aufgaben mit GCS Sync.

`cloud_mastering_service/` · `WIF OIDC Auth`

DMG Release Package (94 MB)

10

Vollständig signiertes, installierbares Standalone DMG Package inkl. gebündelter ARM64 Binaries.

`auratone_ai_2.0.0_aarch64.dmg` · `Tauri Bundler`

Tauri 2 IPC Commands & Rust Handler

Vollständige Übersicht aller nativen Rust-IPC-Befehle und Kernel-Module

TYPE	COMMAND / IPC ROUTE	BESCHREIBUNG	RUST HANDLER / MODUL
COMMAND	cmd_analyze_track	Audio-Analyse via Hermetischem DSP-Sidecar (BPM & Key)	commands.rs / dsp_supervisor
COMMAND	cmd_process_mastering	Echte LUFS-Normalisierung via FFmpeg Sidecar Engine	commands.rs / pipeline.rs
COMMAND	cmd_generate_peaks	Metal GPU High-Res Peak Extraction (Unified Memory)	metal_peaks.rs
COMMAND	cmd_batch_process	Parallele Batch-Verarbeitung via Tokio Async Queue	batch.rs
COMMAND	cmd_cancel_batch	Laufenden Batch-Job unverzüglich abbrechen	batch.rs / CancellationToken
COMMAND	cmd_store_keychain_secret	Passwort / Token verschlüsselt in macOS Keychain ablegen	keychain.rs
COMMAND	cmd_get_keychain_secret	Verschlüsseltes Token aus macOS Keychain lesen	keychain.rs
COMMAND	cmd_scan_directory	Audio-Verzeichnis scannen & SQLite Katalog abgleichen	db.rs / catalog.db
COMMAND	cmd_export_traktor_nml	Traktor Pro 4 NML Playlists & Track-Marker generieren	commands.rs / catalog.db
COMMAND	cmd_sync_apple_music	Apple Music Playlists via JXA osascript Bridge lesen	apple_music.rs
COMMAND	cmd_get_db_metrics	System- & Datenbank-Statistiken abfragen	db.rs / history.db

RUST KERNEL CORE MODULES (SRC-TAURI / SRC /)

- main.rs (App Init & Setup)
- db.rs (3-Way SQLite Triade Engine)
- pipeline.rs (Ingest Pipeline)
- dsp_supervisor.rs (Sidecar Process Manager)
- metal_peaks.rs (Metal Shaders)
- batch.rs (Tokio Queue)
- keychain.rs (Security Framework)
- watcher.rs (FSEvents)

Client-Module & Versionshistorie

React 19 Frontend Modulübersicht, Chunks & Entwicklungsmeilensteine

SRC / SERVICES / & COMPONENTS

tauriBridge.ts — Native Tauri IPC Wrapper & Fallback
TraktorWaveform.tsx — Metal GPU Waveform Canvas
CamelotWheel.tsx — Interactive 1A-12B Harmonic Wheel
DeckSection.tsx — Dual-Deck DJ Player Controls
auratoneMastering.ts — Client Mastering Controller
harmonicEngine.ts — Camelot Key Distance & Scoring
curatorAgent.ts — Gemini 1.5 Pro Musicology Sync
audioPlaybackEngine.ts — Web Audio API Core
traktorExport.ts — NML / M3U / CSV Writer

SRC / LIB / & STORES

camelot.ts — Camelot Key System Rules & Translation
energyScore.ts — Track Energy Level Calculator
localLibraryStore.ts — Zustand Store for Catalog
profileStore.ts — User Profiles & LUFS Targets
errorToastStore.ts — Global Toast Notification Bus
onboardingStore.ts — Guided App Tour State
analytics.ts — Privacy-Preserving Diagnostics
traktorTheme.ts — Color Tokens for Key & Frequency

VERSIONSHISTORIE

- v2.0.0** August 2026
Native macOS Standalone Release. Tauri 2 Rust Kernel, Metal GPU Waveforms, Hermetic Python DSP Sidecar & 3-Way SQLite Triade. 100% Verifiziertes DMG Package (94 MB).
- v1.3.0** 25. Juni 2026
Production Release (Express 4.21 Node.js). 10-Punkte Plan: Error Handling, Profile, Analytics, Single-Endpoint Ingest, Apple Music Integration, Auto-Updates.
- v1.2.1 / v1.2.0** Ende 2025
Dual-Deck DJ Player, KI-Curator, Gemini Transitions, Camelot Wheel, Traktor NML Import/Export Stabilität.
- v1.0.0** 9. Juni 2026
Erster Production Release. Ersatz für PN10 & MIK11: Librosa Key/BPM, FFmpeg Mastering, Harmonische Set-Sortierung.

PROPRIETÄRE SOFTWARE – ALLE RECHTE VORBEHALTEN

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Empirische Benchmarks & SLSA Level 3

Gemessene DSP-Laufzeiten, Hardware-Sicherheit & Kryptografische Provenance

Realer Track (5:18 Min) — Warm

14.1s

Median über Läufe 2–10 (p95: 15.7s). Vollständige BPM & Camelot Key Extraktion.

Cold Start Initialisierung

56.6s

Lauf 1 inklusive PyInstaller Shared Library & Numba JIT Warmup.

Synthetischer 30s Snippet

1.4s

Ultra-fast Preview Detection für Live-Curation (p95: 1.5s).

Voller Master-Durchlauf

52.8s

Analyse → FFmpeg Filterkette → Re-Analyse & LUFS Verifikation.

SECURITY & SUPPLY CHAIN PROVENANCE (SLSA LEVEL 3)

Zero Static Keys

GCP Workload Identity Federation (WIF) OIDC. Keine statischen Keys.

SLSA Level 3

Offizieller slsa-github-generator Workflow für kryptografische Herkunft.

Keychain Vault

Verschlüsselte Aufbewahrung von Auth-Tokens im macOS Keychain Vault.

AuraTone AI v2.0.0 System Architecture & Build Verification Complete

Designed & Architected by Kevin Kuck · Lead System Architect & Senior AI Engineer
Bridging Apple Silicon Metal Hardware with Autonomous AI Reasoning.